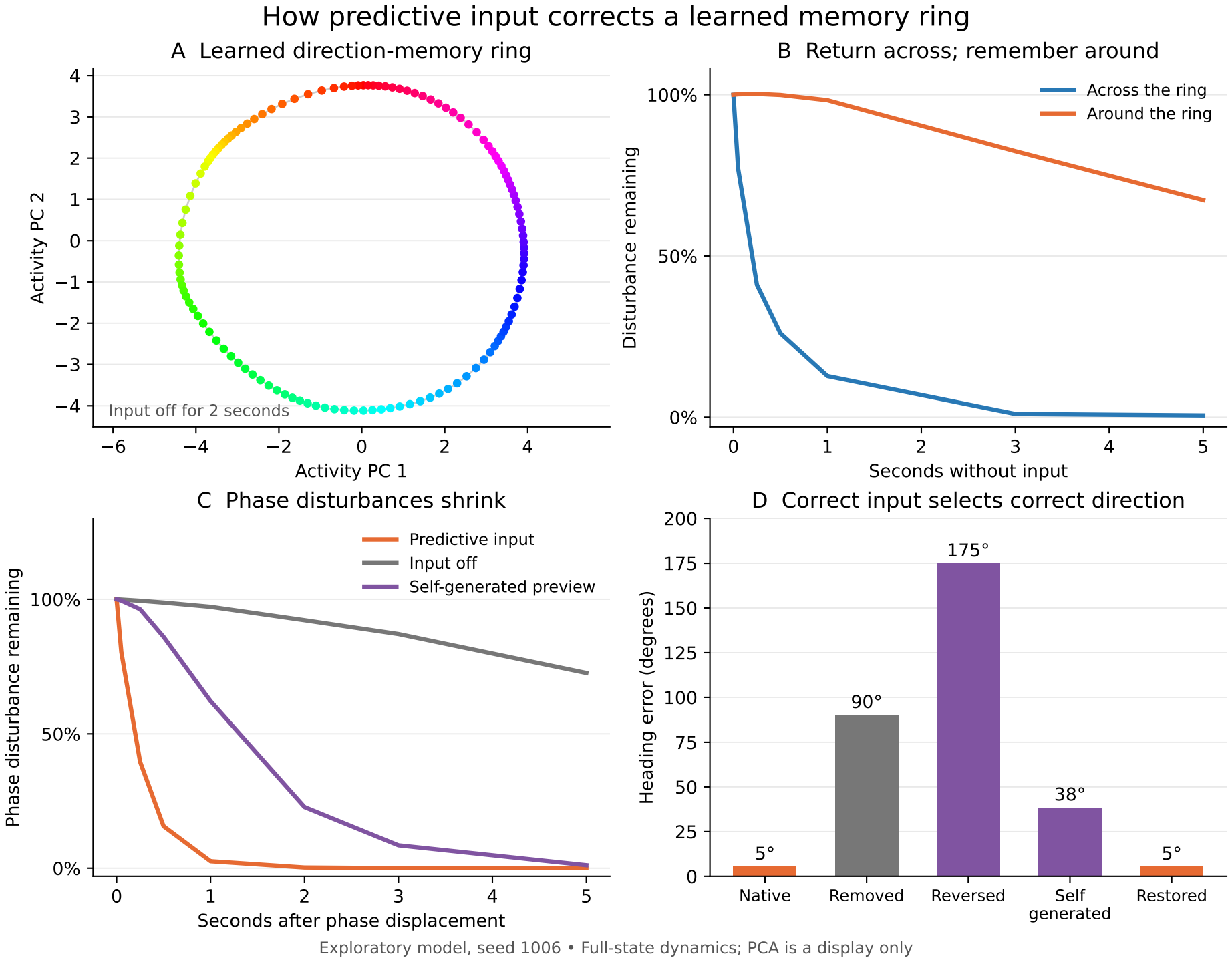
How predictive input corrects a learned memory ring
A Learned direction-memory ring
Activity PC 2
Activity PC 1
Input off for 2 seconds
B Return across; remember around
Across the ring
Around the ring
Disturbance remaining
100%
50%
0%
Seconds without input
C Phase disturbances shrink
Predictive input
Input off
Self-generated preview
Phase disturbance remaining
100%
50%
0%
Seconds after phase displacement
D Correct input selects correct direction
Heading error (degrees)
5°
90°
175°
38°
5°
Native
Removed
Reversed
Self generated
Restored
Exploratory model, seed 1006 • Full-state dynamics; PCA is a display only